



MOHSEN MARZBAN

MSc ICT for Smart Societies · Instrumentation & Electrical Engineer · IoT · ML · Industry 4.0
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SUMMARY

Final-year M.Sc. student in ICT for Smart Societies at Politecnico di Torino, with 6+ years of hands-on engineering experience in industrial instrumentation, electrical systems, and automation. Combines deep operational expertise with academic training in Machine Learning, IoT system design, Python, Docker, cloud platforms, and smart systems. Uniquely positioned to bridge Operational Technology (OT) and Information Technology (IT), with strong practical and analytical skills applicable to Industry 4.0, smart energy, embedded systems, illness UPDRS diagnosis and data-driven engineering roles.

TECHNICAL SKILLS

Programming	Python, MATLAB, SQL, Bash/Linux scripting
ML & Data	Scikit-learn, TensorFlow, NumPy, Pandas, Matplotlib, regression, classification, model evaluation, feature engineering, data visualization
IoT & Embedded	Raspberry Pi, Arduino, MQTT, BLE, LoRaWAN, REST APIs, sensors/actuators, real-time data pipelines, edge computing, Mosquitto MQTT Broker, Local & Global Networks (LAN/WAN), NetBird VPN, Digital twins.
Cloud & DevOps	Docker, AWS fundamentals, Linux (scripting, process management), Git/GitHub, CI/CD basics, UI/UX for dashboards
Databases	SQL Server, MySQL, MongoDB, InfluxDB
Tools & Platforms	Jupyter Notebook, Grafana, Power BI, AutoCAD 2D/3D, Autodesk Navisworks, Microsoft Office Suite
Industrial / OT	P&ID, DCS/F&G, SCADA concepts, instrumentation calibration & testing, hi-pot/megger testing, as-built drawings, PMBOK.
Domain Knowledge	Smart grids, smart buildings & BMS, smart mobility, GIS & satellite positioning, applied cryptography & security fundamentals, communication networks, illness (e.g: Parkinson) UPDRS diagnosis

EDUCATION

M.Sc. — ICT for Smart Societies

Politecnico di Torino | Turin, Italy | 2023 – Present (Final Year)

Relevant coursework: Statistical Learning & Neural Networks, Mobile & Sensor Networks, ICT in Building Design, ICT for Smart Mobility, Communication & Network Systems, Smart Grids, Big Data Systems, Operational Research, Applied Cryptography.

B.Sc. — Electrical Engineering

Government-National University | Gorgan, Iran | 2015 – 2017

Associate Degree — Electrical Engineering

Government-National University | Yazd, Iran | 2013 – 2015

PROFESSIONAL EXPERIENCE

Instrumentation & Electrical Technician

Maadkoush Steel / Behin Tarh Paydar / Farsnov Cement | Bandarabbas & Shiraz, Iran | Oct 2020 – Jul 2024

- Procured electrical and instrumentation components; quality assurance against industry standards.
- Conducted thorough inspections and functional tests on purchased field devices, verifying compliance with technical specifications and sector regulations.
- Coordinated with suppliers for construction and repair of specific product-line sections, maintaining project schedules and documentation.

Electrical & Instrumentation Technical Office Engineer

Faradast Energy Falat — PGSOC (World's Largest Gas Condensate Refinery) / Control Poyan |

Bandarabbas, Iran | Jan 2018 – Jan 2020

- Analysed complex engineering documents: P&IDs, DCS & F&G systems, loop diagrams, electrical schematics, layout drawings, technical datasheets, inspection reports, and punch lists.
- Installed and commissioned field instrumentation: transmitters (TT, TE), pressure/level switches (PS, LS), indicators (LG, PG, TG), and valves (gate, globe, ball, butterfly).
- Calibrated temperature transmitters (TT) to verify measurement accuracy and traceability.
- Junction Box (JB) wiring: ensured correct terminations per loop diagrams, layout specifications, and cable schedules.
- Electrical installations: commissioned LV switchboards, motor starters (DOL), lighting systems (SLD drawings, fixture installation), and cable laying (underground UG and above-ground AG).
- Performed hi-pot and megger tests on cables to certify insulation integrity and electrical safety.
- Created and maintained As-Built drawings for lighting and instrumentation systems.

SELECTED ACADEMIC PROJECTS

IoT & ML Systems — Smart Societies Coursework

Politecnico di Torino | Turin, Italy | 2023 – Present

- IoT-based monitoring system: designed sensor/actuator pipelines using MQTT and InfluxDB for real-time data ingestion, dashboarding (Grafana), and automated alerts.
- Machine Learning projects: applied regression, classification, and neural network models (Scikit-learn, TensorFlow) to structured datasets; performed cross-validation, feature engineering, and ROC analysis.
- Smart grid & energy systems: analysed distributed energy architectures, demand-response strategies, and grid optimization scenarios.
- Smart mobility data analysis: worked with timestamped telemetry data for system optimization and predictive modelling tasks.
- Cloud & DevOps: containerised applications using Docker, integrated CI/CD pipelines with GitHub, and deployed to cloud environments (AWS basics).

LANGUAGES

Persian	Native
English	Proficient — C1 (Listening, Reading, Writing, Speaking)
Italian	Intermediate — B1 (Listening, Reading, Writing, Speaking)

VALUE PROPOSITION

I bring a rare combination: years of real-world engineering practice in high-stakes industrial environments (oil & gas, steel, cement) fused with an MSc curriculum designed for tomorrow's smart, connected world. I can read a P&ID and write a data pipeline. I understand both the sensor on the factory floor and the cloud infrastructure processing its data — making me immediately productive in Industry 4.0, IoT, energy, Mobility and automation roles.